🐍 Day 1 - Python Overview

Welcome to **Day 1** of the **Python 45 Days Training Program**.

Today's goal is to understand what Python is, why it is popular, where it is used, and how to write your first Python program.

📖 Learning Objectives

By the end of this session, you will be able to:

- Understand what Python is.
- Know the history of Python.
- Explain why Python is popular.
- Identify Python applications.
- Install Python and Visual Studio Code.
- Write and execute your first Python program.

📚 What is Python?

Python is a **high-level**, **interpreted**, and **general-purpose programming language**. It is designed to be easy to read and write, making it one of the best programming languages for beginners and professionals.

Python emphasizes **code readability** using simple syntax that resembles the English language.

👨‍💻 History of Python

- Created by **Guido van Rossum**.
- Development started in **1989**.
- First released in **1991**.
- Named after the comedy television show **Monty Python's Flying Circus**.
- Maintained by the **Python Software Foundation (PSF)**.

⭐ Features of Python

- Easy to Learn
- Simple Syntax

- Open Source
- Cross Platform
- Interpreted Language
- Object-Oriented
- Large Standard Library
- Dynamically Typed
- Automatic Memory Management
- Supports Multiple Programming Paradigms

🚀 Why Learn Python?

Python is one of the most in-demand programming languages because it is used in many industries.

Popular Applications

- 🌐 Web Development
- 🤖 Artificial Intelligence
- 🧠 Machine Learning
- 📊 Data Science
- 📈 Data Analytics
- ☁ Cloud Computing
- 🔐 Cyber Security
- ⚙ Automation
- 🖥️ Desktop Applications
- 🎮 Game Development
- 🌍 Internet of Things (IoT)

🏢 Companies Using Python

Many leading companies use Python in production.

- Google
- Microsoft
- Netflix
- Instagram
- Spotify
- Dropbox
- Reddit
- Uber
- Amazon
- NASA

🆚 Why Python Over Other Languages?

Python	C	Java
Easy Syntax	Complex	Moderate
Less Code	More Code	More Code
Interpreted	Compiled	JVM Based
Beginner Friendly	No	Moderate
Large Libraries	Limited	Good

Example:

C

```c
#include<stdio.h>

int main()
{
    printf("Hello World");
    return 0;
}
```

Python

```python
print("Hello World")
```

Python requires fewer lines of code, making development faster and easier.

🛠️ Installing Python

Download Python from the official website:

https://python.org

During installation:

- Select **Add Python to PATH**
- Click **Install Now**

Verify installation:

```
python --version
```

or

```
py --version
```

💻 Installing Visual Studio Code

Download VS Code:

https://code.visualstudio.com

Install the following extension:

- Python Extension (Microsoft)

📂 Python File Extension

Python programs are saved using the `.py` extension.

Example:

```
hello.py
```

▶ Running a Python Program

Using Command Prompt or Terminal:

```
python hello.py
```

or

```
py hello.py
```

🎉 Your First Python Program

```
print("Hello, Python!")
```

Output

```
Hello, Python!
```

🖨️ Understanding `print()`

The `print()` function is used to display output on the console.

Example:

```
print("Welcome to Python")
print("My Name is Pavan")
print("Today is Day 1")
```

Output:

```
Welcome to Python
My Name is Pavan
Today is Day 1
```

🧪 Practice Programs

Program 1

```
print("Hello Python")
```

Program 2

```
print("My Name is John")
```

Program 3

```
print("I am learning Python Programming")
```

Program 4

```
print("Python is Easy")
print("Python is Powerful")
print("Python is Fun")
```

📝 Hands-on Activity

Create a file named `introduction.py`.

Write a program to display:

- Your Name
- College Name
- City
- Favorite Programming Language
- Career Goal

Example:

```
print("Name : Rahul")
print("College : Presidency University")
print("City : Bangalore")
print("Favorite Language : Python")
print("Career Goal : Software Engineer")
```

📌 Best Practices

- Use meaningful file names.
- Save files with the `.py` extension.
- Keep code properly indented.
- Write clean and readable code.
- Practice daily.

❓ Interview Questions

1. What is Python?
2. Who created Python?
3. When was Python released?
4. Why is Python popular?
5. What is the file extension of Python?
6. Is Python compiled or interpreted?
7. Name five applications of Python.
8. Which function is used to display output?
9. Is Python open source?
10. Name three companies that use Python.

🏠 Assignment

Task 1

Write a Python program to display:

```
Hello Python

My Name is _______

I am learning Python.

Python is used for AI.

Python is used for Automation.

Thank You.
```

Task 2

Research and write short notes on:

- Python Software Foundation
- Guido van Rossum
- Latest stable version of Python
- Five real-world applications of Python

🎯 Day 1 Summary

Today you learned:

- Introduction to Python
- History of Python
- Features of Python
- Advantages of Python
- Applications of Python
- Installing Python
- Installing VS Code
- Running Python Programs
- `print()` Function
- Writing Your First Python Program

🚀 Next Session

Day 2 – Python Installation, IDE Setup, Variables, and Input/Output Basics

Happy Coding! 🐍